

Symmetry protection on the lattice — audit and next-step research report

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1. Current-state assessment of the thesis

Verdict: the QFT and goofy chapters are sound reviews, but the lattice chapter — the thesis's only original result — does not yet support its headline claim. The "symmetric" data are not symmetric, and no number carries an uncertainty.

Question, as reconstructed. Can 't Hooft's criterion (a parameter is natural if setting it to zero enlarges the symmetry) be made visible both in a non-perturbative lattice RG and in one-loop QFT? If so, the hierarchy problem reads as "no symmetry forbids H^+H^- ", and goofy reads as a candidate for that missing symmetry.

Logical chain. (1) The RG preserves symmetries. (2) Ising: Z₂-odd couplings stay zero under blocking unless Z₂ is broken. (3) QFT: $\delta m \propto m$ for fermions, $\delta m\phi^2 \propto m\psi^2$ for scalars. (4) Hierarchy problem = no symmetry for H^+H^- . (5) Goofy might supply one.

Claim in the thesis	Where	Status	Reason
One-loop fermion mass shift $\delta m \propto m$	§3.1, App. B.1	Correct (textbook)	Re-derived; divergent part $3g^2m/(16\pi^2)\cdot(2/\epsilon)$ checks.
One-loop scalar shift $\delta m\phi^2 \propto y^2m\psi^2$	§3.2, App. B.2	Correct; sign convention inconsistent	With $m_0^2 = m^2 + \delta m^2$ the stated condition gives $\delta m^2 = -\Sigma(0)$. Physics unaffected.
Fermion divergence "only logarithmic"	§5	Not shown	Dimensional regularization hides power divergences; the Λ^2 vs $\ln \Lambda$ contrast needs a cutoff calculation.
Odd couplings stay ≈ 0 when $h_0 = 0$	§2.3, Fig. 2	Not established	The $h_0 = 0$ chain starts all-up and stays magnetized ($m \approx 0.2\text{--}0.5$ for all 300 samples); one chain, no error bars.

Claim in the thesis	Where	Status	Reason
Odd couplings grow more for larger breaking	§2.3, Fig. 2	Not established	$h(b=8)/h_0 = 367$ for $h_0 = 1e-4$ but 49 for $h_0 = 1e-3$: no linear response, unknown noise.
h_3 drift at $h_0 = 0$ is truncation + flat loss direction	§2.3, App. A.6	Incorrect as argued	For Z_2 -symmetric data the convex pseudo-likelihood has its unique minimum at $h = h_3 = 0$, whatever the even truncation. Flatness changes variance, not bias.
Breaking through h alone populates the whole odd sector	§5	Not established	h_3 also drifts at $h_0 = 0$; the gap at $b = 8$ (0.012 vs 0.03) has no uncertainty.
The Ising study builds intuition for goofy	§1, §2.1, §5	Analogy only	Z_2 is an ordinary internal symmetry protecting an odd operator; goofy targets an even mass operator.
Goofy forbids the Higgs mass, RG-stable to all orders	§4	Literature claim	Checked against the cited papers in §12.

What can be inferred today. The pipeline runs end to end, and in one realization $h(b=8)$ is about ten times larger at $h_0 = 1e-3$ than at $h_0 = 0$. That is qualitatively consistent with Z_2 protection, but it is not yet a measurement.

What the central claim really is. The defensible claim is "symmetry protects operators under RG": standard in QFT, and a theorem for a Z_2 -covariant blocking, so the lattice numerics test the pipeline rather than the physics. The stronger reading, that Ising builds intuition for how goofy protects the Higgs mass, has no support in the current work.

The most important unanswered question. The lattice chapter shows only the protected side. The hierarchy problem is about an unprotected relevant operator, and the same Ising model contains one: the energy operator $\varepsilon \sim \phi^2$, whose coupling must be tuned to reach criticality. The thesis never shows what "unprotected" looks like on the lattice.

Smaller issues a referee would flag.

- Appendix A.5 says h was initialized at $1e-3$ in every run; the notebooks initialize it at h_0 itself (0, $1e-5$, $1e-4$, $1e-3$, $5e-3$).

- "300 equilibrium configurations": 200 are used, 30 Metropolis sweeps apart, from an all-up start; equilibrium is never checked.
- Eq. (A.6) keeps 4 of the 12 local-field terms that the three-spin Hamiltonian (A.5) produces, so the inferred h_3 is not the coupling of any Hamiltonian.
- Naturalness is credited to 't Hooft through a review [5]; cite 't Hooft (1980) directly.
- §1.1 "RG preserves symmetries" needs two caveats, a symmetric regulator and no anomaly. Both matter for goofy.

2. Current-state assessment of the repository

Verdict: the code is a faithful extension of Di Carlo's pipeline, and its loss and gradients are correct. But the sampler never reaches the ensemble the thesis assumes, the simulated point is not critical, and the three-spin field is not a Hamiltonian gradient.

Structure. `src/IsingRG_3spin.py` holds the Metropolis sampler with h and h_3 , majority-rule blocking, and the pseudo-likelihood loss with analytic gradients; `src/IsingRG.py` is an import shim. There is one template notebook, five executed Colab notebooks (`notebooks/runs/h0_*`, one per h_0), the thesis figures and the manuscript. The multi-run overlays of thesis Figs. 2 and 3 are not produced by any file in the repository.

Pipeline as run. $L = 240$, $K = (0.203, 0.078)$, all spins up, 300 Metropolis samples 30 sweeps apart, first 100 dropped. The 200 kept configurations are blocked at $b = 2, 3, 4, 5, 6, 8$ and fitted by plain gradient descent ($\eta = 0.025$, 7000 steps) in the basis $\{K_1-K_4, h, h_3\}$.

Finding	Evidence	Consequence
The "near-critical" point is paramagnetic	Binder cumulant at (0.203, 0.078): $U_4 = 0.588, 0.558, 0.510, 0.341$ for $L = 30, 60, 120, 240$, falling with L (critical value 0.611). Crossing at $K_{1c} = 0.2059(2)$ for $K_2 = 0.078$.	$K_1 = 0.203$ sits 1.4 % below criticality; at $L = 240$ the correlation length is smaller than the box. Di Carlo's paper does not quote this point; it comes from his example notebook.
The $h_0 = 0$ chain never samples a symmetric ensemble	All runs start all-up; the stored $m(t)$ stays between 0.2 and 0.5 for all 300 samples. 6000 sweeps is far shorter than the Metropolis magnetization autocorrelation time here.	The "Z2-symmetric" data are a single-sign, non-equilibrium ensemble.

Finding	Evidence	Consequence
One chain per h_0 , all with the same seed	Every run notebook uses <code>jax.random.PRNGKey(1)</code> .	No run-to-run spread exists, so differences between h_0 values carry no uncertainty.
The three-spin field is not a gradient	Θ_i keeps 4 of the 12 terms of $\partial O_3 / \partial \sigma_i$. Mixed derivatives differ ($\partial \Theta_i / \partial \sigma_j = 2$, $\partial \Theta_j / \partial \sigma_i = 0$ for $j = i + x$). A single flip changes O_3 by $-2\sigma\Phi$ in 50 of 50 random tests, by $-2\sigma\Theta$ in 30 of 50.	The inferred h_3 is the coefficient of a non-Gibbs conditional model. Forward Monte Carlo with $h_3 \neq 0$ would break detailed balance (latent: $h_3 = 0$ in every run).
Fixed-step gradient descent, no convergence test	Notebook cells; thesis Fig. 5 shows the $h_0 = 5 \times 10^{-3}$ fit unconverged after 75 000 steps. The odd couplings start at $h = h_0$, not at 10^{-3} as Appendix A.5 says.	Results depend on the step budget, although the loss is concave with a unique optimum.
Loss and gradients are correct	An independent implementation reproduces the original loss to 10^{-14} ; the original analytic gradient equals JAX autodiff.	The pseudo-likelihood machinery can be reused as is.
Citation metadata wrong	README and CITATION.cff title Di Carlo's arXiv:2401.04811 "Coarse-graining the Ising model with the inverse Ising problem". The paper is Monte Carlo, blocking, and inference: How to measure the renormalization group flow .	Easy fix.

Quantities available but never analysed. Per-sample magnetization (it reveals the stuck chain), run-to-run spread, the curvature of the loss (standard errors), a criticality check of the simulated point, and the even-sector flow against Di Carlo's fixed point.

Reproduction. The original JAX sampler, run here, reproduces the slow relaxation from the all-up state ($m = 0.64, 0.58, 0.59, 0.57$ for the first four samples, about 15 s per sample). For speed I re-implemented the identical random-site Metropolis dynamics in numba and added a Wolff cluster sampler. Both agree with exact enumeration on a 4×4 lattice, and Wolff reproduces the exact critical Binder value 0.611 at $K_c = 0.4407$ for $L = 30$ to 240.

3. Main scientific weaknesses and gaps

The central gap: the thesis demonstrates (imperfectly) only the protected half of naturalness on the lattice, while the hierarchy problem is about the unprotected half — and the same Ising model contains it. Ranked by severity:

1. **No valid null test.** The $h_0 = 0$ data are a single-sign, non-equilibrium sample, and nothing has an error bar. "Odd couplings stay at zero" is therefore not yet measured, and the h_3 offset is unexplained.
2. **The lattice statement is a theorem, so the numerics test the pipeline, not the physics.** Majority-rule blocking commutes with $\sigma \rightarrow -\sigma$. For Z2-symmetric data the pseudo-likelihood F obeys $F(\theta_{\text{even}}, \theta_{\text{odd}}) = F(\theta_{\text{even}}, -\theta_{\text{odd}})$, and F is strictly concave, so its unique optimum has $\theta_{\text{odd}} = 0$ for any truncation of the even sector. A non-zero odd coupling at $h_0 = 0$ therefore needs an asymmetric sample, not a flat loss direction. Physics content can only come from quantitative features: the response to explicit breaking, the relevance of h , and a contrast with an unprotected direction.
3. **The broken runs do not measure an RG property.** The inferred $h(b=8)$ equals $367 \cdot h_0$ at $h_0 = 10^{-4}$ and $49 \cdot h_0$ at $h_0 = 10^{-3}$, which no linear flow allows. The stored magnetizations (0.56 and 0.64 at $b = 2$) differ more than the fields do; §9 tests whether the sample magnetization, not h_0 , drives the odd couplings.
4. **The unprotected side is missing.** The Ising model is the lattice version of Z2 ϕ^4 theory: $\sigma \sim \phi$, $h \sim a \phi$ source, and the energy operator $\epsilon \sim \phi^2$ is the scalar mass. Z2 forbids h but allows ϵ , so reaching criticality (a massless scalar) needs tuning of K . That tuning is the Ising model's own hierarchy problem, and the thesis never shows it.
5. **The goofy analogy is aimed at the wrong operator.** Z2 forbids an odd operator, as discrete chiral symmetry forbids $\bar{\psi}\psi$. Goofy must forbid an even bilinear, $H^\dagger H$, which every linear unitary internal symmetry leaves invariant. That is exactly why goofy has to be unconventional (it flips the kinetic term), and why no Ising-type internal symmetry can mimic it. A meaningful lattice analogue must protect the even operator ϵ .
6. **The QFT chapter is scheme-blind.** In dimensional regularization there is no Λ^2 ; the hierarchy problem appears as sensitivity to heavy masses ($\delta m \phi^2 \propto y^2 m \psi^2$), which the thesis computes correctly. The "quadratic versus logarithmic" language of the conclusions needs a cutoff calculation, which is the bridge to the lattice. Also, in $\overline{\text{MS}}$ $\beta(m^2) \propto m^2$ holds in the SM by dimensional analysis alone, so the RG stability of $m^2 = 0$ is not what makes goofy non-trivial.
7. **The goofy review overstates its sources.** De Boer, Goertz and Incrocci call kinetic-term invariance (it needs imaginary spacetime coordinates) a speculation; the thesis presents the full SM action as goofy-symmetric. Trautner states that finite

contributions are not forbidden and that the goofy-breaking scale should lie near the electroweak scale; the thesis says the hierarchy problem is "addressed without fine-tuning". Details in §12.

4. Candidate future directions

Six candidates were considered: the four in the brief, and two that the audit itself suggested (E1, E2).

- **A. Repair and deepen the Ising study.** Equilibrium sampling, error bars, a correct three-spin operator, an exact optimizer; then quantify protection: null tests, linear response to h_0 , and the RG eigenvalue of h .
- **B. Multi-component spin model with a goofy-like transformation.** Two Ising species σ , τ per site; the natural " $H \rightarrow iH$ " analogue is the rotation $\sigma \rightarrow -\tau$, $\tau \rightarrow \sigma$ of $\psi = \sigma + i\tau$.
- **C. Complexified Ising model.** Let $\sigma \rightarrow i\sigma$ act on the Boltzmann weight.
- **D. Strengthen the perturbative QFT side.** Cutoff versus dimensional regularization, counterterm structure, and the one-loop RG stability of the 2HDM relation $m_{11}^2 + m_{22}^2 = 0$.
- **E1. The unprotected direction in the same model.** The thermal operator $\varepsilon \sim \phi^2$ is Z_2 -even, relevant and allowed: measure its RG eigenvalue, its "radiative" shift, and the tuning it requires.
- **E2. Duality protection of the even operator.** Kramers–Wannier (KW) self-duality maps $\varepsilon \rightarrow -\varepsilon$. Test whether duality-preserving interactions leave the critical point unshifted while duality-breaking ones move it.

What the audit says about B and C before any ranking.

- C is not a statistical-mechanics model. For Ising spins, $\sigma \rightarrow i\sigma$ sends every pair coupling $K \rightarrow -K$ (the antiferromagnet, an ordinary model) and $h \rightarrow ih$: complex weights and the Lee–Yang problem, with no positive measure to sample and no link to mass protection.
- Majority-rule blocking does not commute with the staggering that relates K and $-K$. So even the formal coupling-space map has no RG covariance on the lattice.
- B, in its goofy-like form, reduces to an ordinary symmetry. For equal couplings the rotation $\sigma \rightarrow -\tau$, $\tau \rightarrow \sigma$ is the Z_4 of the Ashkin–Teller model. It forbids odd and mixing terms, but leaves the even energy operator $\sigma\sigma + \tau\tau$ invariant, which is exactly the operator goofy must forbid.
- The only known lattice mechanism that protects an even relevant operator is self-duality. In the Ashkin–Teller form, that is E2 with heavier infrastructure (a two-species cluster algorithm).

5. Comparative evaluation and ranking

Ranking: E1 and A first (they are two halves of one measurement), E2 a close and cheap third, D a useful appendix, B and C last. The 14 criteria of the brief are grouped into six; each group is scored 0–3.

Direction	Relevance to protection and RG (criteria 1, 4, 5)	Link to goofy (6)	Genuine result, real conclusion (3, 11, 14)	Rigor and clarity (2, 7)	Feasibility and reuse of code (8, 9, 10)	Fit for a master's thesis (12, 13)	Total
E1. Unprotected even direction	3	2	2	3	3	3	16
A. Repair, quantify the odd sector	3	1	2	3	3	3	15
E2. KW duality protection	2	3	2	3	3	2	15
D. QFT: cutoff, counterterms, 2HDM	2	2	1	3	3	3	14
B. Multi-component spins	1	1	1	2	1	1	7
C. Complexified Ising	0	1	0	0	1	1	3

Why the scores fall this way.

- A is necessary but not sufficient. Without it nothing on the lattice is defensible, yet on its own it only re-proves a theorem with error bars.
- E1 wins on substance. It shows, in the same simulation and the same RG, a relevant operator that is not protected. That is the hierarchy problem's lattice form, and it uses the existing pipeline with no new model.
- E2 is the only route that touches goofy's actual mechanism: a transformation that is a symmetry of the RG flow but not of the Boltzmann weight, and that forbids an even mass operator. It reproduces known physics (O'Brien–Fendley, Seiberg–Seifnashri–Shao) in a new framing, so it can support an analogy, not evidence for goofy.
- D clarifies the Wilsonian versus dimensional-regularization picture at almost no cost, but yields no new result.
- B collapses to an ordinary Z_4 symmetry, which cannot forbid the even operator, unless it becomes the Ashkin–Teller version of E2 at higher cost. C is ill-defined (§4).

Decision. One direction: make the Ising study a quantitative test of naturalness on both sides — a protected relevant operator (h) against an unprotected one (ε) in the same lattice RG (A + E1). E2 is added as a bounded bridge to goofy, and D as a one-page appendix.

6. Chosen direction and justification

Chosen: protected versus unprotected relevant operators in one lattice RG. The 2D Ising model has two relevant operators: the odd magnetic operator (h , exact $y_h = 15/8$) and the even thermal operator ($\varepsilon \sim \phi^2$, exact $y_t = 1$). The question is which of them needs tuning, and why.

The question it answers. Does Z2 make h multiplicatively renormalized (generated only in proportion to the explicit breaking, the lattice form of $\delta m \propto m$) while ε , though less relevant, is additively renormalized and must be tuned (the lattice form of $\delta m^2 \propto \Lambda^2$)? The bridge then asks whether an unconventional transformation — KW self-duality, a symmetry of the RG flow but not of the Boltzmann weight — can protect ε .

Why the thesis does not answer it yet. Its lattice chapter shows only the protected side, without a valid null test, and never mentions ε .

Why it beats the alternatives.

- It needs no new model: the same Hamiltonian, sampler family and blocking.
- It is calibrated: both exponents are known exactly, so the method itself is tested.
- It is falsifiable: each claim has a null test that could fail (§7).
- It maps one-to-one onto the hierarchy problem: a relevant operator with a symmetry needs no tuning; one without a symmetry does.

What can realistically be shown. Odd couplings vanish within errors in ergodic data and exactly in symmetrized data. The thesis's offsets come from its single-sign sample. The linearized RG is block-diagonal in Z2 parity with $y_h \approx 1.875$ and $y_t \approx 1$. The critical coupling is shifted by fluctuations by order one in lattice units and depends on every even coupling, while $h_c = 0$ for all of them. KW-preserving interactions leave the transition at $g = 1$; KW-breaking ones shift it linearly.

New insight it adds. Relevance creates sensitivity; symmetry decides whether that sensitivity becomes a tuning problem. The more relevant operator ($y_h = 15/8$) needs no tuning, while the less relevant one ($y_t = 1$) does. The Higgs mass operator is the worst case: the most relevant scalar operator ($y = 2$ in 4D) with no ordinary symmetry able to forbid it.

Risks, stated before running.

- Inferred odd couplings in magnetized samples may be biased ("leakage"), which could obstruct reading y_h off the inferred-coupling flow. Mitigation: MCRG for exponents.
- Truncation of operator bases, finite-size effects, and noise in linear-response estimates.
- Over-reading the KW analogy as evidence for goofy. Mitigation: label it an analogy throughout.

Why it suits a master's thesis. It reuses the thesis's code and narrative, needs hours of laptop compute, and ends in a conclusion rather than a list of future work.

7. Detailed research plan

Eight steps, each with a control that could fail. Steps 0–6 use the Ising pipeline; step 7 is the duality bridge.

Setup. $H = -\sum_d K_d S_d - h M - h_3 O_3$ on an $L \times L$ torus, with S_d the pair sums at Manhattan distance d and O_3 the sum of L-shaped triplets. Two microscopic points: the nearest-neighbour model at the exact $K_c = 0.440687$ (critical, self-dual), and the thesis model $K = (0.203, 0.078)$. RG map: $b \times b$ majority rule, ties broken by a fair coin, which commutes with $\sigma \rightarrow -\sigma$ in distribution.

Step	Question	Method	Control or null test	Success criterion
0. Validation	Are sampler and inference right?	Exact enumeration (4×4), Binder cumulant at exact K_c , $b = 1$ recovery of couplings, fluctuation–dissipation identity $A(b=1) = 1$	Exact values	Agreement within errors; any failure stops the study
1. Criticality	Is $(0.203, 0.078)$ critical?	Binder crossings, $L = 30\text{--}240$, Wolff	NN model at K_c must give 0.611	Crossing located to 0.1 %
2. Null test	Do odd couplings vanish when Z_2 is exact?	Inverse Ising at $b = 2\text{--}8$ on ergodic Wolff ensembles, jackknife errors	Symmetrized data must give exactly 0	$ pull < 3$ at every b , no trend in b

Step	Question	Method	Control or null test	Success criterion
3. Thesis offsets	Where do the $h_0 = 0$ offsets come from?	Thesis protocol, 6 seeds; equilibrium halves with $m > 0$ and $m < 0$; larger even bases	Full ensemble versus single-sign halves	Offsets appear in halves, vanish in the full ensemble
4. Explicit breaking	Is the odd sector multiplicatively renormalized?	Inferred couplings at $h_0 = 10^{-5}$ – $3 \cdot 10^{-3}$, $L = 60$ and 120	Odd couplings odd in h_0 , even couplings even	Plateau of θ_{odd}/h_0 at $h_0 \cdot \chi \ll 1$, crossover near $h_0 \cdot \chi \approx 1$
5. Linearized RG	Are y_h, y_t reproduced; is T block-diagonal?	MCRG with iterated 2×2 majority rule, $L = 240$ and 256	Even-odd blocks of T must be zero; exact $15/8$ and 1	y_h within 1 %, y_t within 3 %, cross-block pulls of order one
6. Unprotected direction	Does ϵ need tuning?	K_c versus K_2 ; mean-field versus exact K_c ; correlation length at the thesis point	$h_c = 0$ for all K_2 by symmetry	K_c moves with every even coupling by order one
7. Duality bridge	Can KW protect ϵ ?	Exact diagonalization, $N = 8$ – 16 , crossings of $N \cdot \Delta\sigma$	Pure TFIM must give $g_c = 1$	$g_c = 1$ within 10^{-3} for KW-symmetric couplings, linear shift for KW-breaking ones

What would count as failure. Step 2 fails if raw odd couplings sit beyond three standard errors with a trend in b : that would indicate an asymmetric sampler or blocking. Step 4 fails if θ_{odd}/h_0 is not constant at small $h_0 \cdot \chi$. Step 7 fails if g_c moves with the KW-symmetric coupling.

Statistics. Wolff configurations are stored 2 cluster sweeps apart, with integrated autocorrelation times measured. Errors come from a delete-one jackknife over 20 contiguous blocks, checked against independent replicas. Seeds and parameters are recorded in every output file.

Systematics. Operator-basis truncation (bases of 6, 7 and 11 couplings; MCRG truncations of 1–5 even and 1–4 odd operators), finite size ($L/b \geq 15$), blocking parity (odd and even b), and the convergence of the optimizer (exact Newton steps).

Figures needed. (a) Magnetization traces, thesis protocol versus Wolff. (b) Odd couplings versus b with errors, the replacement for thesis Fig. 2. (c) Pure-state leakage versus basis. (d) Field response θ_{odd}/h_0 versus $h_0 \cdot \chi$. (e) MCRG exponents by level. (f) K_c versus K_2 , exact against mean field. (g) KW crossings g_c versus coupling.

8. Work actually completed

All eight steps of §7 were carried out on a 2-core machine, with new code in the repository (branch `extension-symmetry-protection`); nothing in the original code was overwritten.

New code.

File	Purpose
<code>src/isingflow/lattice.py</code>	Shell sums, the correct three-spin field $\Phi = \partial O_3 / \partial \sigma$ (12 terms), the thesis field Θ kept for comparison, plaquette field, operator totals
<code>src/isingflow/mc.py</code>	Numba Metropolis (same dynamics as the original) and Wolff cluster sampler with the field included by cluster-flip acceptance
<code>src/isingflow/inference.py</code>	Majority-rule blocking, features, exact Newton optimizer for the pseudo-likelihood, jackknife errors, fluctuation-dissipation response
<code>src/isingflow/mcrg.py</code>	Swendsen MCRG: 5 even and 4 odd operators, iterated 2×2 majority rule, eigen-exponents with jackknife
<code>src/isingflow/kwchain.py</code>	Exact diagonalization of KW-self-dual qubit chains in Z_2 parity sectors
<code>tests/test_pipeline.py</code>	Validation suite (§10)
<code>scripts/*.py</code> , <code>scripts/queue_*.sh</code>	Every run below, with seeds and parameters written into each output file
<code>RESEARCH_LOG.md</code>	Chronological log of tests and findings

Runs.

Run	Model and size	Statistics	Purpose
Binder scans	$K_2 = 0.078$, $K_1 = 0.195$ – 0.211 ; NN at K_c ; $L = 30$ – 240	4000–6000 Wolff sweeps per point	Criticality of the thesis point; $K_c(K_2)$
Thesis protocol	$(0.203, 0.078)$, $L = 240$, $h_0 = 0, 10^{-4}, 10^{-3}$	6 seeds $\times$ 200 kept Metropolis samples	Reproduce the thesis and its run-to-run spread
Equilibrium, thesis point	Same couplings, $h_0 = 0$ (two replicas), $10^{-4}, 10^{-3}$	4000 ($h_0 = 0$) and 2000 Wolff configurations	Null test, single-sign leakage, field response
Equilibrium, critical NN	$K_c = 0.440687$, $L = 240$ and 256 , $h_0 = 0$	4000 and 8000 configurations	Null test at criticality; MCRG
Field series	NN at K_c , $L = 60$ and 120 , $h_0 = 0, 10^{-5} \dots 3 \cdot 10^{-3}$ (7 values)	3000 configurations each	State channel versus field channel
Second critical model	$K_2 = 0.078$ at its K_c , $L = 256$	8000 configurations	Universality of the MCRG exponents
KW chains	$H = -\sum(ZZ + gX) + \lambda \cdot \text{KW-even} + \kappa \cdot \text{KW-odd terms}$, $N = 8$ – 16	Exact	Duality protection of ε

Settings shared by all inverse-Ising fits. Majority rule with fair-coin ties; bases $\{K_1$ – $K_4, h, h_3\}$ with Θ (thesis) or Φ (corrected), plus enlarged bases with the plaquette and shells 5–8; Newton steps to a decrement below 10^{-15} per site; 20 contiguous jackknife blocks; Wolff configurations 2 cluster sweeps apart (measured autocorrelation time of m about 0.5 samples).

Theory added. A proof that inferred odd couplings vanish for symmetric data at any truncation; the single-sign leakage mechanism; the one-loop cutoff calculation behind the "quadratic versus logarithmic" claim; and a one-loop check showing that the goofy mass relation of the 2HDM is not preserved by a positive cutoff.

9. Numerical and theoretical results

Z2 protection now holds as a measurement: odd couplings vanish in ergodic samples and exactly in symmetrized ones, and the linearized RG splits into an odd block ($y_h = 1.8748(5)$, exact $15/8$) and an even block ($y_t = 0.99(1)$, exact 1). The thesis offsets are

leakage from its single-sign sample. The even, mass-like operator is unprotected and needs tuning, unless a duality forbids it.

9.1 Theory

T1. Inferred odd couplings vanish for symmetric data, at any truncation. Under $s \rightarrow -s$ the pseudo-log-likelihood obeys the identity below. With Z2-symmetric data the objective is even in θ_{odd} , and strict concavity makes the optimum unique, so $\theta_{\text{odd}} = 0$ whatever even operators are kept. A flat loss direction changes only the variance; it cannot bias an odd coupling, so thesis App. A.6 has the cause wrong.

$$\ell(\theta_e, \theta_o; -s) = \ell(\theta_e, -\theta_o; s) \implies F(\theta_e, \theta_o) = F(\theta_e, -\theta_o) \implies \theta_o^* = 0$$

T2. Single-sign leakage. A sample confined to $m > 0$ has the same local conditionals as the symmetric ensemble (DLR), so a complete basis still gives $\theta_{\text{odd}} = 0$. A truncated basis does not: the misfit of the even sector acquires a mean $\propto m$ in a magnetized sample, and the constant and three-spin features absorb it, $\theta_{\text{odd}} \approx c(b, \text{basis}) \cdot m_b$.

T3. Cutoff versus dimensional regularization (one loop). With a hard cutoff Λ the fermion shift stays $\propto m$ and logarithmic, while the scalar shift is additive and quadratic. The logarithm's coefficient, $3y^2 m_\psi^2 / 4\pi^2$, matches the thesis's $1/\epsilon$ pole; dimensional regularization simply drops the Λ^2 term that the lattice keeps.

$$\delta m_\psi = \frac{3g^2 m_\psi}{16\pi^2} \ln \frac{\Lambda^2}{m_\psi^2} + \dots, \quad \delta m_\phi^2 = -\frac{y^2}{4\pi^2} \left[\Lambda^2 - 3m_\psi^2 \ln \frac{\Lambda^2}{m_\psi^2} + 2m_\psi^2 \right] + \dots$$

T4. A positive cutoff breaks the goofy mass relation of the 2HDM. Under the goofy (r0) relations $\lambda_1 = \lambda_2$ and $\lambda_6 = -\lambda_7$, the one-loop quadratic divergences of m_{11}^2 and m_{22}^2 have identical scalar and gauge parts, so their sum shifts by twice that amount instead of staying zero. The relation survives only in mass-independent schemes, consistent with Trautner's remark that the cutoff must flip sign under the exotic space-time map. No lattice can realize goofy protection of this relation.

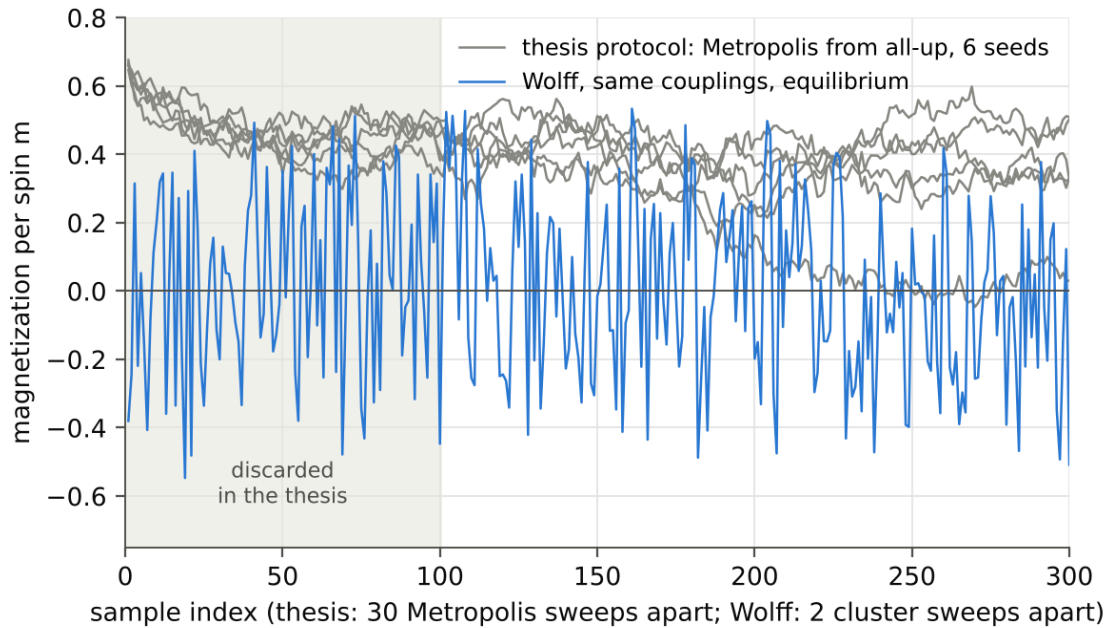
$$\delta(m_{11}^2 + m_{22}^2) \Big|_{\lambda_1=\lambda_2, \lambda_6=-\lambda_7} = \frac{2\Lambda^2}{16\pi^2} (3\lambda_1 + 2\lambda_3 + \lambda_4) + (\text{gauge, Yukawa}) \neq 0$$

9.2 The thesis protocol, reproduced

Six independent runs of the exact thesis protocol show that the $h_0 = 0$ chains never leave $m > 0$, while an ergodic chain at the same couplings samples both signs.

The thesis's $h_0 = 0$ chains never leave $m > 0$; an ergodic chain visits both signs

$L = 240$, $K = (0.203, 0.078)$, $h_0 = 0$



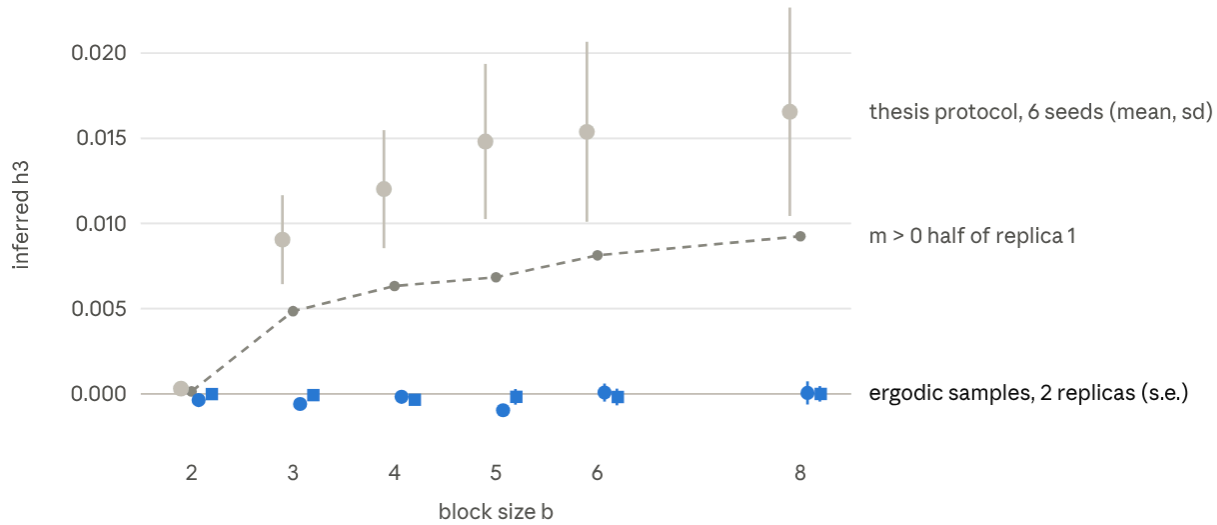
h_0	$h(b=8)$, thesis	$h(b=8)$, 6 reproductions (mean $\pm$ sd)	$h_3(b=8)$, thesis	$h_3(b=8)$, 6 reproductions
0	-0.004	+0.003 $\pm$ 0.012	0.012	0.017 $\pm$ 0.006
10^{-4}	0.037	0.004 $\pm$ 0.008	0.029	0.017 $\pm$ 0.007
10^{-3}	0.049	0.047 $\pm$ 0.028	0.031	0.029 $\pm$ 0.004

$h_0 = 10^{-4}$ cannot be told apart from $h_0 = 0$ at any b under this protocol, and the thesis's $h(8) = 0.037$ lies about four standard deviations above the reproductions. $h_0 = 10^{-3}$ is distinguishable from $b = 4$ on. Gradient descent and the exact optimum agree to within 6 %, so the optimizer was not the problem; the sampling was.

9.3 Null test and leakage at $h_0 = 0$

At $h_0 = 0$, h_3 vanishes in ergodic samples; the thesis offset is single-sign leakage

Inferred three-spin coupling h_3 (thesis field) vs block size; $L = 240$, $K = (0.203, 0.078)$



results/thesis_protocol.npz and prot_thesis_L240_h0*.npz · 6 Metropolis seeds; two Wolff replicas of 4000 configurations; thesis three-spin field

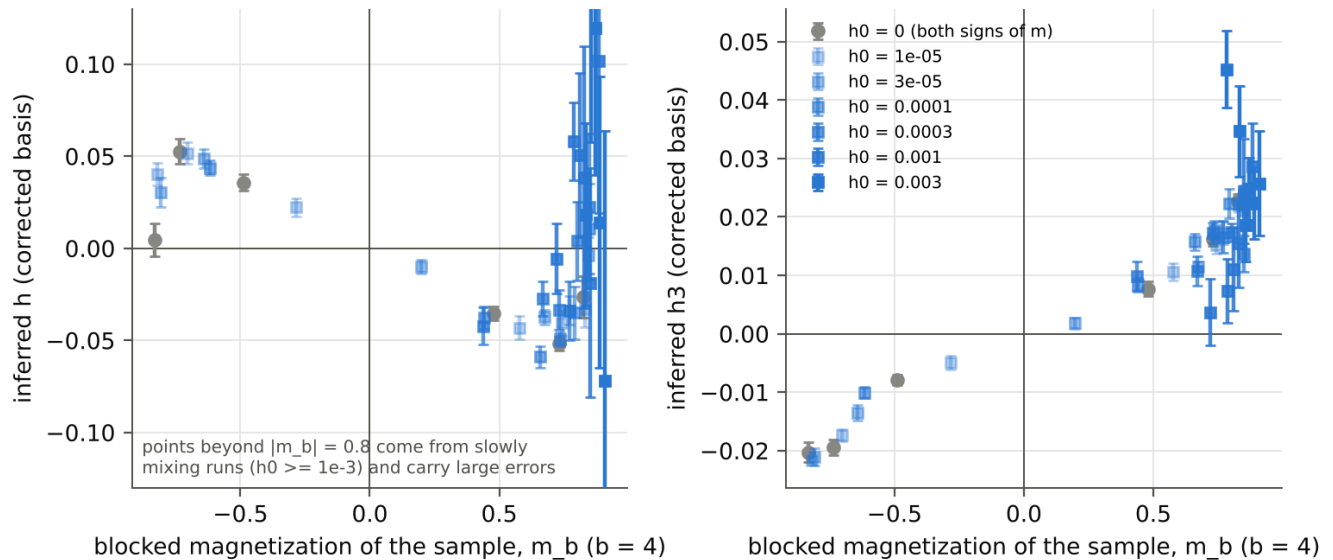
Over $b = 2-8$ and both bases, the raw odd couplings give rms pulls of 1.7 and 0.9 in two independent replicas at the thesis point and 1.2 at the critical NN point. The largest pull, 3.6, appears in one replica only; split-half and replica comparisons (§10) show these errors are accurate to 10–30 %. Symmetrized data give $|\theta_{\text{odd}}| < 3 \times 10^{-17}$. At $b = 1$, where the basis contains the true Hamiltonian, even the single-sign halves give h and h_3 within 1.7 standard errors of zero, which confirms that the leakage is a truncation effect.

At $b = 8$ the leakage is $h_3 = +0.009$ (thesis field) for $m_b = 0.34$ at the thesis point and $+0.023$ for $m_b = 0.69$ at criticality. Enlarging the even basis (plaquette, shells 5–8, an infinite-range term) does not remove it.

9.4 Explicit breaking: the state channel dominates

Inferred odd couplings follow the sample's magnetization, whatever the field h_0

NN Ising at K_c , $L = 120$; six magnetization bins per data set; h_0 from 0 to 3×10^{-3}

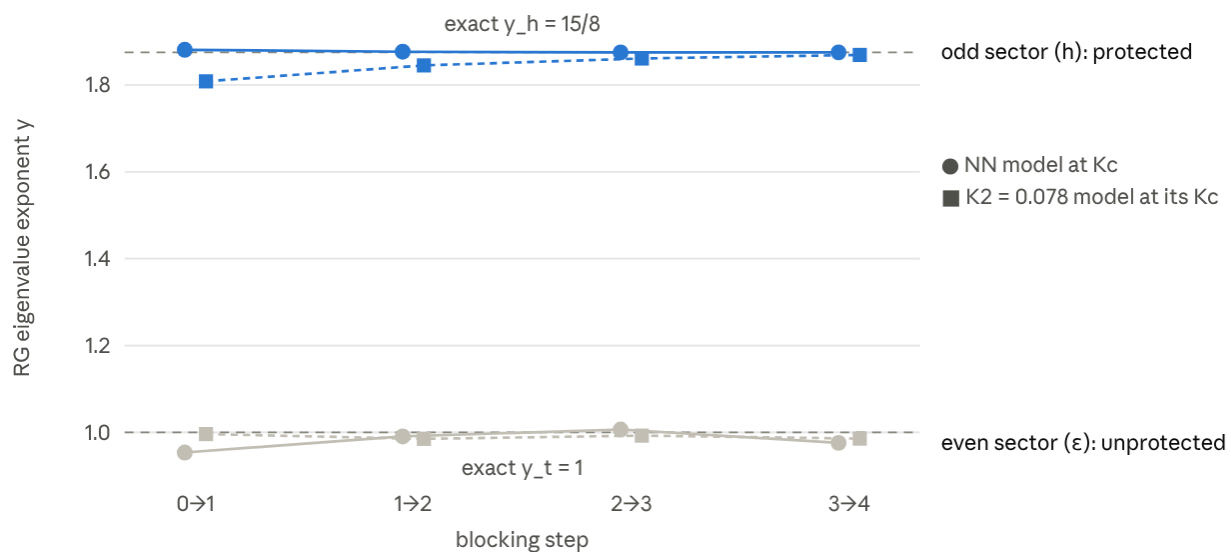


Across 42 magnetization bins of 7 data sets ($h_0 = 0$ to 3×10^{-3} , critical NN model, $L = 120$), the inferred odd couplings fall on one odd function of the sample magnetization, whatever h_0 . Fitting $\theta_{\text{odd}} = c \cdot m_b + R \cdot h_0$ gives a state coefficient of up to 0.06 per unit m_b , while the field coefficient R is not resolved (5 to 48 per unit h_0 , with errors as large as the values). The thesis's broken-symmetry flows are therefore not measurements of the RG eigenvalue of h .

9.5 Linearized RG: block-diagonal, one relevant direction per sector

Two relevant directions, in two different models: $y_h \rightarrow 15/8$ (odd), $y_t \approx 1$ (even)

Leading eigenvalue exponents of the MCRG matrix per 2×2 majority-rule step; $L = 256$, 8000 Wolff configurations each



results/mcrg_nn_L256.npz and mcrg_k2c_L256.npz · 8000 Wolff configurations each; statistical jackknife errors

The even-odd cross blocks of the MCRG matrix are consistent with zero (rms pulls 0.7–1.5 per step). Two different microscopic models approach the same exponents and share the same scaling fields, which is universality made visible. The left eigenvectors below (normalized to their largest component) are the scaling fields as combinations of block couplings.

Scaling field	nn	diag	x2	plaq	knight	m	tri	line3	star5
magnetic, $\gamma = 1.875$	−0.001	−0.001	−0.001	0.000	−0.001	0.305	1	0.497	0.201
thermal, $\gamma = 0.98$	0.261	0.346	0.459	0.148	1	0.000	−0.001	0.002	0.001

The magnetic field has no component on any even operator, so no change of the even couplings can generate h . The thermal field has a component on every even operator, so every change of the even couplings moves the system off criticality.

9.6 The unprotected direction needs tuning

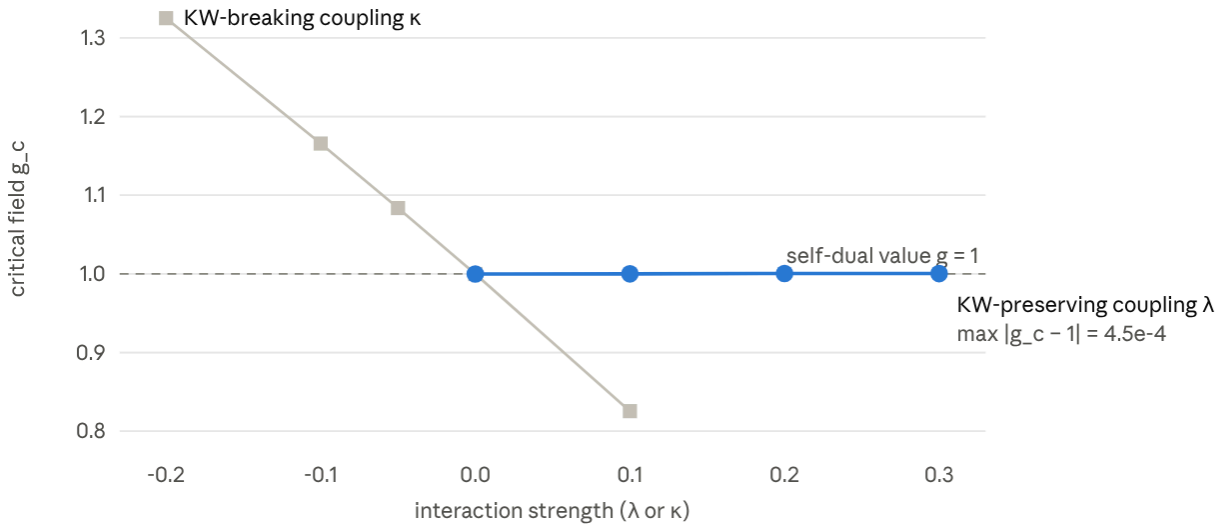
K2	K1c, this work	Mean field ($4K1 + 8K2 = 1$)	Fluctuation shift
0	0.44069 (exact, self-dual)	0.250	+0.191
0.04	0.3157(5)	0.170	+0.146
0.078	0.2059(2)	0.094	+0.112

Fluctuations move the critical coupling by 0.11–0.19 in lattice units, the lattice counterpart of a cutoff-sized mass correction, and the critical line has slope -3.0 against the mean-field -2 . At the thesis point, 1.4 % below $K1c$, the correlation length is about 76 lattice spacings. A Barbieri–Giudice measure $\Delta = |\partial \ln \xi / \partial \ln K2| \approx 80$ there, roughly ξ/a , is the Ising analogue of $(\Lambda/m_H)^2$. The odd direction needs no tuning: $hc = 0$ for every $K2$.

9.7 Duality protects the even operator

Self-duality pins the transition at $g = 1$; breaking it moves g_c linearly

Critical field of $H = -\sum(ZZ + gX) + \text{interaction}$, exact diagonalization $N = 8\text{--}16$, extrapolated



results/kw_scan.npz · exact diagonalization, crossings of $N \cdot \Delta\sigma$ for $N = 8\text{--}16$, linear extrapolation in $1/N^2$

KW-preserving couplings up to $\lambda = 0.3$ keep the extrapolated critical field within 5×10^{-4} of 1 ($N = 8\text{--}16$). The KW-breaking coupling moves it linearly, with slope $dg_c/d\kappa = -1.7$, so with it one must retune g . This is the same contrast as Z2 versus the thermal direction, now for the even, mass-like operator.

10. Robustness checks and possible failure modes

Every headline number survived the checks below. Three limitations remain and bound what this report claims: the field channel of the inverse-Ising flows is unresolved, the quoted errors are accurate only to 10–30 %, and the MCRG exponents carry statistical errors only.

Validation of the tools.

Check	Result	Verdict
Wolff and Metropolis against exact enumeration (4×4 at K_c)	$\langle m^2 \rangle$ and $\langle m^4 \rangle$ agree to < 0.3 %	Pass
Binder cumulant at the exact K_c , $L = 30\text{--}240$	0.607(4), 0.613(2), 0.610(3), 0.613(2); exact 0.61069	Pass
Couplings recovered without blocking ($b = 1$, $L = 48$)	All six within 1.4σ of (K_c , 0, 0, 0, 0, 0)	Pass

Check	Result	Verdict
Fluctuation–dissipation identity $A(b=1) = 1$	Diagonal 0.96–1.54, largest pull 2.4	Pass, low precision
Symmetrized data	Odd couplings below 3×10^{-17}	Pass (exact)
Original JAX loss and gradient at the new optimum	Loss equal to 10^{-14} ; original gradient 3×10^{-12}	Pass
Parity-sector Hamiltonians against dense ED ($N = 8$)	Identical spectra	Pass

The suite (tests/test_pipeline.py) passes in 41 s. It caught one bug before any production run: a Wolff driver that stopped each interval at a target cluster volume, a state-dependent stopping time, gave a Binder cumulant of 0.63 instead of 0.61. All data use a fixed number of cluster moves per interval.

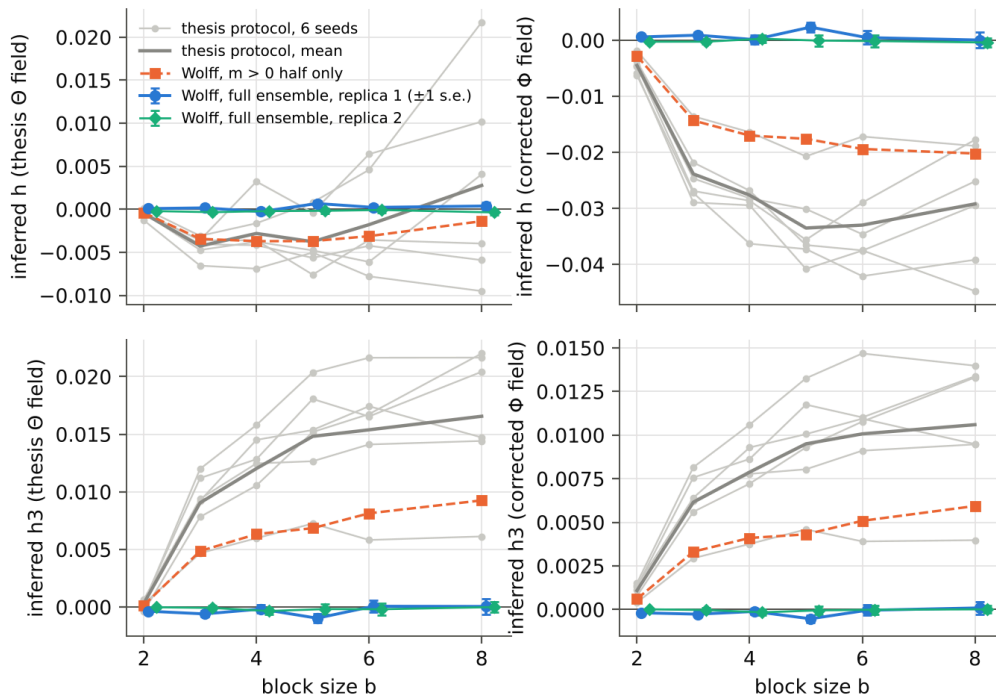
Statistical errors.

- **Autocorrelation.** At $h_0 = 0$ every data set has $\tau_{\text{int}}(m) \approx 0.5$, $\tau_{\text{int}}(|m|) \leq 1.1$ and $\tau_{\text{int}}(E) \leq 1.5$ saved configurations, far below the 150–400 configurations in a jackknife block. A field slows the Wolff update: $\tau_{\text{int}}(E)$ reaches 32–316 for $h_0 \geq 3 \times 10^{-4}$ at $L = 120$, and 74 at the thesis point with $h_0 = 10^{-3}$. Those sets hold only tens of effective samples, so their errors are underestimated. They enter only §9.4, where they could hide a field signal but not create one.
- **Calibration.** If the jackknife errors are right, the difference of two independent estimates has an rms pull of 1. Split halves of each chain give 0.93 for the odd couplings and 1.28 for the even ones; the two independent replicas give 1.37 and 0.99. Odd-coupling errors are accurate to 10–30 %; even-coupling errors are about 25 % too small.
- **The null test at that precision.** Pooled over three data sets, the raw odd pulls have an rms of 1.3. The largest, 3.6, belongs to a coherent offset in replica 1 at $b = 2\text{--}5$ ($h_3 \approx -2 \times 10^{-4}$ in the Φ basis), where the $m < 0$ configurations leak slightly more than the $m > 0$ ones. The offset is absent in replica 2 and in both critical data sets, and it is 50–100 times smaller than the thesis offsets. Sampler and blocking are exactly symmetric in distribution, so it can only be a fluctuation: the null test passes at about 1.3 times the jackknife errors. At the critical point of the thesis model ($K_1 = 0.2059$, $L = 256$, 8000 configurations), the raw pulls at $b = 4$ and 8 lie between 1.2 and 1.8.

Systematics.

At $h_0 = 0$ the odd couplings vanish in the ergodic ensemble; the thesis offsets track the sample's sign

$L = 240$, $K = (0.203, 0.078)$; exact pseudo-likelihood optimum; left: thesis three-spin field Θ , right: corrected Φ



Check	Result
Single-sign halves without blocking ($b = 1$, $L = 120$)	h and h_3 within 1.6σ of zero in both halves and both bases: the leakage needs a truncated basis
Leakage against the even basis ($m > 0$ half; net odd local field $h + h_3(x_3)$ at $b = 4 / b = 8$)	Critical NN model: 0.047 / 0.070 with four shells, 0.053 / 0.081 with the plaquette, 0.052 / 0.080 with shells 5–8, 0.047 / 0.075 with an infinite-range term, 0.053 / 0.086 with all of them. Thesis point: 0.013–0.016 / 0.022–0.029. No enlargement removes it
Thesis field Θ against Φ	Same null-test outcome; Θ puts the leakage into h_3 , while Φ splits it between h and h_3 with opposite signs
Leakage at criticality in two models	The same coefficient: Θ -basis $h_3 = 0.023$ at $m_b \approx 0.68$ ($b = 8$) in both the NN and the K2 model, which suggests that the fixed point and the truncation, not microscopic details, set the leakage
MCRG operator truncation	y_h stable to 10^{-3} with 1–4 odd operators; y_t needs at least two even operators (one gives 0.90–0.97; two to five agree within 0.005)

Check	Result
MCRG lattice size and model	NN at $L = 240$ and 256 agree ($y_h = 1.8752(5)$ and $1.8746(3)$ at step $2 \rightarrow 3$). y_t scatters between 0.954 and 1.007 over steps, so it is reproduced to $\pm 3\%$, not $\pm 1\%$. The K2 model approaches $15/8$ from below ($1.808 \rightarrow 1.869$ in four steps) and has not fully converged; the last step uses 16×16 lattices
Critical line	K1c($K2 = 0.04$) refined from a coarse-grid interpolation, $0.3175(10)$, to $0.3157(5)$ (crossings at 0.3162 for $L = 60/120$ and 0.3157 for $120/240$); §9.6 is corrected. The fluctuation shift (0.146) and the slope (-3.0) are unchanged
KW finite size	Adding $N = 18$ moves every extrapolated g_c by at most 1.4×10^{-4} ; the self-dual estimates stay within 4×10^{-4} of 1 (0.99990 to 1.00037) against KW-breaking shifts of 0.08 – 0.33 . The crossing search once picked noise-level crossings deep in the ordered phase ($\lambda = 0.3$, small N); it now skips points where both gaps are below 10^{-6}

Failure modes, and what they would mean.

Possible failure	Status	Consequence had it occurred
Sampler non-ergodic or biased	Excluded: exact checks, critical Binder value, $\tau_{\text{int}}(m) \approx 0.5$ at $h_0 = 0$	Null test meaningless
Blocking not Z2-covariant	Excluded: fair-coin ties; symmetrized data exactly zero	Spurious odd couplings
Optimizer not converged	Excluded: Newton decrement below 10^{-15} per site	Results depend on the step budget
Leakage read as RG physics	Identified and explained (T2)	Thesis Figs. 2–3 misread

Possible failure	Status	Consequence had it occurred
Field channel unresolved	Open: R reaches 3–5 σ only for h_3 at the thesis point, from four data sets with slow mixing at $h_0 = 10^{-3}$	No inverse-Ising measurement of y_h ; MCRG is used instead
Simulated point not critical	Confirmed for the thesis point; critical runs added	Thesis flows are paramagnetic crossovers, not critical flows

11. Physical interpretation

On the lattice the results are direct demonstrations. Their QFT reading is a strong analogy that rests on universality. The hierarchy reading is a heuristic, and the link to goofy is a structural analogy only.

Statement	Level	Status	Basis
Blocking never generates odd couplings unless Z2 is broken, explicitly or by the sample's state	Lattice	Direct demonstration	T1; null tests; symmetrized data exactly zero
The linearized RG splits into an odd and an even block, each with one relevant direction	Lattice	Direct demonstration	MCRG for two microscopic models, y_h and y_t reproduced
The thermal operator is unprotected, and its critical value depends on every even coupling	Lattice	Direct demonstration	$K_c(K_2)$ against mean field; components of the thermal scaling field
Self-duality pins the transition; breaking it moves the transition	Lattice (quantum chain)	Direct demonstration of a known result	Exact diagonalization; O'Brien–Fendley
h is a ϕ source, ε the ϕ^2 mass term, and tuning K is additive mass renormalization	QFT/RG	Strong analogy (universality)	Continuum limit of Z2 ϕ^4 theory; exponents reproduced

Statement	Level	Status	Basis
KW protection of ε is chiral protection of a Majorana mass	QFT/RG	Exact mapping in 2D	Jordan–Wigner; KW is a one-site Majorana translation
Tuning grows as $(\xi/a)^{\gamma_t}$; for the Higgs, $\gamma = 2$ gives $(\Lambda/m_H)^2$	Hierarchy	Heuristic (naturalness criterion, here quantified)	$\Delta \approx 80$ against $\xi \approx 76$ at the thesis point
KW is a lattice counterpart of goofy	Hierarchy/goofy	Suggestive analogy, not evidence	Shared structure, one decisive difference (§12)

Lattice: what is demonstrated, and what is not. Protection of the odd sector is now measured, not assumed: it holds within errors in ergodic data, exactly in symmetrized data, and as a zero block in the linearized RG. The odd sector does respond to explicit breaking in proportion to the breaking, but that proportionality is established by MCRG ($h' = b^{\gamma_h} \cdot h$, with $h = 0$ an invariant hyperplane), not by the inverse-Ising flows. Those are dominated by the sample's magnetization (§9.4), so they cannot be read as the RG response to h_0 . The new and stronger result is the unprotected side: the thermal scaling field overlaps every even operator, so any change of the short-distance couplings moves the system off criticality, and the critical coupling sits 0.11–0.19 away from its mean-field value.

QFT/RG: universality makes the dictionary precise. The 2D Ising critical point is the continuum limit of $Z_2 \phi^4$ theory, with $\sigma \sim \phi$ ($\Delta = 1/8$) and $\varepsilon \sim \phi^2$ ($\Delta = 1$). The lattice's two relevant directions are therefore the ϕ source (forbidden by Z_2) and the ϕ^2 mass term (allowed). The non-universal, cutoff-dependent position of the critical point is the lattice form of additive mass renormalization: the bare value at which the renormalized mass vanishes is set by short-distance physics, not by symmetry. Two caveats keep the analogy honest. First, in 2D the ϕ^4 tadpole is logarithmic, so what carries over to 4D is the structure (an additive shift that depends on the cutoff), not the power of Λ ; the power follows from the RG eigenvalue, $m/\Lambda \sim |K - K_c|^{(1/\gamma)}$. Second, in its fermionic description 2D Ising is a free Majorana fermion and ε is its mass. In that language KW acts as a one-site Majorana translation, which flips the sign of the mass, so the KW protection of ε is the lattice version of the chiral protection in thesis §3.1, not a new mechanism for scalars.

Hierarchy: relevance creates sensitivity, symmetry decides whether it costs tuning. For a relevant operator with eigenvalue γ , the sensitivity of the correlation length to the bare couplings grows as $\Delta \sim (\xi/a)^{\gamma}$. At the thesis point, $\Delta \approx 80$ for $\xi \approx 76$ lattice spacings ($\gamma_t = 1$). For the Higgs mass operator, $\gamma = 2$ gives the familiar $\Delta \sim (\Lambda/m_H)^2$. The magnetic direction is more relevant ($\gamma_h = 15/8$) yet costs nothing: $h = 0$ is selected by symmetry,

and the even couplings can vary freely without generating it. This is the hierarchy problem in complete form — a relevant operator with no protecting symmetry — computed in one model with no free parameters. The scaling relation is exact RG; calling large Δ unnatural is the naturalness criterion, which is a judgement rather than a theorem.

12. Implications for the GOOFY / hierarchy-problem motivation

The extension strengthens the motivation for looking beyond ordinary symmetries, but not the case for goofy itself. On one lattice it shows that the obstacle goofy targets is real and measurable, and that a non-ordinary transformation can remove it. It gives no evidence that goofy does so for the Higgs, and one calculation (T4) sharpens a caveat already present in the goofy papers.

Why this is more than an illustration of protection. The original lattice chapter illustrated the textbook case: an ordinary internal symmetry protecting an odd operator. That case is not in question, and it is not what goofy is about. Two results go further.

- **The obstacle, measured.** In the same RG, Z_2 leaves the even, mass-like operator ϵ unprotected, and reaching criticality costs a tuning that grows like $(\xi/a)^{\gamma_t}$. The 4D statement is general: every linear unitary internal symmetry leaves $H^\dagger H$ invariant, so no symmetry of that kind can forbid it. Status: direct demonstration on the lattice; a standard group-theory fact in 4D.
- **An unconventional protection exists.** KW self-duality is not an ordinary internal symmetry — on the lattice it is non-invertible (Seiberg–Seifnashri–Shao) — yet it protects ϵ exactly and non-perturbatively. Status: direct demonstration of a known result.

How far the KW parallel goes.

Feature	KW duality (Ising chain)	Goofy (2HDM, SM)
Action on the mass operator	flips it: $(g - 1) \rightarrow -(g - 1)$	flips it: $m^2 H^\dagger H \rightarrow -m^2 H^\dagger H$
Action on the nearest-neighbour or kinetic term	exchanges ZZ and X	flips the sign of $(DH)^\dagger (DH)$
Map on coupling space that commutes with the RG	yes; the self-dual locus $g = 1$ is RG-invariant	yes; an outer automorphism with an RG-fixed hyperplane (de Boer–Trautner)
Exact symmetry of the regulated theory on the protected locus	yes, of the lattice Hamiltonian at $g = 1$	only with $\delta \rightarrow -i\delta$ and $A \rightarrow -iA$, which de Boer, Goertz and Incrocci call speculative

Feature	KW duality (Ising chain)	Goofy (2HDM, SM)
Survives a positive cutoff	yes, by construction	no: the cutoff must flip sign, $\Lambda^2 \rightarrow -\Lambda^2$ (Trautner; T4)

The first three rows are the structure that de Boer and Trautner formalize: a transformation of coupling space under which the beta functions are covariant fixes an RG-invariant hyperplane, and the symmetry of the beta functions can exceed that of the action. KW is a lattice instance of that structure. The last two rows are where the analogy stops: KW is a symmetry of the regulated theory, and goofy, so far, is not.

What T4 adds. With a hard positive cutoff, the one-loop quadratic divergences of m_{11}^2 and m_{22}^2 are equal under the goofy relations $\lambda_1 = \lambda_2$, $\lambda_6 = -\lambda_7$, so $m_{11}^2 + m_{22}^2 = 0$ shifts by twice that amount instead of staying zero. The relation therefore survives only in mass-independent schemes, which agrees with Trautner's statement that invariance of the effective action requires $\Lambda^2 \rightarrow -\Lambda^2$ in the exotic space-time case. Two consequences follow. No lattice with a positive Boltzmann weight can realize goofy protection of this relation. And what goofy forbids is the Higgs mass counterterm, which makes the mass calculable, while finite contributions from heavy scales can still arise (Trautner says so explicitly); whether they are small is a model-by-model question.

What $H \rightarrow iH$ means. The map $h_i \rightarrow i h_i$ acts on real field components, so it is not a unitary change of variables and not a symmetry of a positive Euclidean weight: it reverses the sign of the kinetic term and makes the weight non-normalizable. For Ising spins the same move, $\sigma \rightarrow i\sigma$, turns a ferromagnet into an antiferromagnet and the field imaginary, which is why the "complexified Ising" direction was rejected (§4). The defensible content of the transformation lies in coupling space: it maps parameters to parameters and leaves the form of the beta functions invariant. That is a precise but limited statement, and it is exactly why no lattice simulation can test goofy directly.

Bottom line. The defensible thesis claim is: the Ising lattice shows what any solution of the hierarchy problem by symmetry must achieve (protect the even, relevant operator), shows that ordinary symmetries fail at it, and shows through KW that an unconventional transformation can succeed in principle. Goofy is a candidate of that unconventional kind whose protection is established for the beta functions in mass-independent schemes and not yet for a physically regulated theory. Claims that it solves the hierarchy problem, or that the full SM action is goofy-symmetric, go beyond its sources (§13).

13. Recommended changes to the thesis

The QFT and goofy chapters need corrected wording and attribution. The lattice chapter needs a new method, new figures and one new section. Changes are listed in order of impact.

Sections to modify or add.

Thesis section	Change	Content
§2.3 Symmetry protection in action	Replace	Wolff sampling, two replicas, jackknife errors, exact Newton optimum; null test with pulls and the symmetrized control; single-sign leakage (T1, T2); MCRG block structure and y_h in place of the inverse-Ising broken flows
New §2.4 The unprotected direction	Add	Thermal exponent and scaling field (MCRG); critical line against mean field; $\Delta \approx 80$ at the thesis point; contrast with $h_c = 0$ for every K2. This is the core new result
§2.1–2.2 Model and setup	Correct	State that (0.203, 0.078) is paramagnetic ($K1c = 0.2059(2)$); replace Θ of Eq. (A.6) by $\Phi = \partial O_3 / \partial \sigma$, or say that h_3 is then the coefficient of a non-Gibbs model
§1, §1.1 Introduction and RG basics	Rewrite, add	State the precise question and the dictionary $\sigma \sim \phi$, $\varepsilon \sim \phi^2$, $h \sim \text{source}$, $t \sim m^2$; add the two caveats to "the RG preserves symmetries" (a symmetric regulator, no anomaly); define y and the tuning measure
§3.1–3.2 Loop calculations	Correct, add	Fix the sign convention of δm^2 ; add the hard-cutoff results (T3), which the "logarithmic versus quadratic" language needs
§3.3 Parallels	Rewrite	$Z_2 \leftrightarrow$ chiral symmetry for the odd case; $\varepsilon \leftrightarrow$ scalar mass for the unprotected case; note that ε is also the Majorana mass

Thesis section	Change	Content
§4.1–4.2 Goofy	Correct, add	Wording corrections below; T4 as a short subsection; attribute RG-stability claims precisely (Ferreira et al.: all orders in the scalar and gauge sectors; Trautner: checked to three loops)
New §4.3 A lattice analogue: KW duality	Add, labelled as analogy	Exact-diagonalization result and the parallels-and-differences table of §12
§5 Conclusions, Abstract	Rewrite	See the proposed conclusion below
App. A.5–A.6	Replace	A.5: odd couplings start at h_0 , not 10^{-3} ; 200 configurations are kept, not 300. A.6: T1 and T2 replace "truncation plus flat direction"; add the $b = 1$ control and the basis table

Claims to weaken, correct or strengthen.

Thesis wording	Replace with	Basis
"such operators are generated under coarse-graining only when the symmetry is explicitly broken" (Abstract)	"For an ergodic Z_2 -symmetric ensemble no odd coupling is generated under coarse-graining, within errors and exactly for symmetrized data; the linearized RG separates a protected odd sector from an unprotected even one"	§9.3, §9.5
"the odd couplings grew under the flow, more strongly for larger breaking" (§5)	"The inferred odd couplings track the sample magnetization rather than h_0 ; the RG response to the field is measured by MCRG, $y_h = 1.875$ "	§9.4, §9.5
"breaking the symmetry through a single operator is enough to populate the entire forbidden sector" (§5)	Keep the statement, prove it with the MCRG magnetic scaling field, which is a fixed mixture of every odd operator	§9.5

Thesis wording	Replace with	Basis
"an artifact of the truncated operator basis rather than a violation of protection" (§5, A.6)	"A single-sign sample combined with a truncated basis; for symmetric data the inferred odd couplings vanish at any truncation"	T1, T2
"the divergence is only logarithmic" (§5)	Keep only with the cutoff calculation, which shows it	T3
"the same contrast we observed on the lattice between the protected and unprotected sectors" (§5)	True only once the new §2.4 exists	§9.6
"offers a natural explanation for why the Higgs mass remains small without requiring fine-tuning" (§1)	"has been proposed as a transformation that could forbid the Higgs mass term"	Trautner; de Boer et al.
"The complete Standard Model action with vanishing Higgs mass then becomes goofy-symmetric" (§4.2)	"Only if the transformation is supplemented by $\delta \rightarrow -i\delta$, $A \rightarrow -iA$, which de Boer et al. describe as speculative; otherwise the kinetic terms break it explicitly"	de Boer et al.; Trautner
"the hierarchy problem is addressed without fine-tuning" (§4.2)	"Both mechanisms make the Higgs mass calculable; the resulting scenarios are strongly constrained, finite contributions from heavy scales can arise, and the goofy-breaking scale must lie near the electroweak scale"	de Boer et al.; Trautner
"the same mathematical footing as conventional symmetry protection" (§4.2)	"The same symmetry-based footing for the RG stability of the relations; it does not make goofy a symmetry of the regulated theory"	de Boer–Trautner; T4

The strengthened claims are the new ones: the unprotected relevant operator and its tuning cost are measured on the lattice, and KW provides an existence proof of protection for an even operator, labelled as an analogy to goofy.

Figures. Replace Fig. 2 with the odd couplings against b with errors (ergodic replicas, thesis protocol, single-sign half). Replace Fig. 3 with the collapse of the odd couplings on the sample magnetization, or drop it. Add the magnetization traces, the Binder test of the

simulated point, the MCRG exponents, the critical line and the KW crossings (all in figures/extension/). Figs. 4–5 (loss landscape, gradient-descent convergence) become unnecessary with an exact optimizer.

New appendix material. Sampler and validation (Wolff with a field, exact checks, autocorrelation, error calibration); proofs of T1 and T2; MCRG method with truncation tables; exact diagonalization of the KW chains; cutoff loop integrals and T4.

References. Cite 't Hooft (1980) directly rather than through a review. Add Swendsen (1979) for MCRG, Wolff (1989) for the cluster algorithm, Kramers and Wannier (1941), O'Brien and Fendley (2018), Seiberg, Seifnashri and Shao (2024), de Boer and Trautner (2026), Trautner's PLB 873 (2026) 140190 for the regulator caveat, Darvishi and Krawczyk (2017) for 2HDM quadratic divergences, and Barbieri and Giudice (1988) for the tuning measure. Correct the Di Carlo title in the repository's README and CITATION.cff.

A conclusion the evidence supports.

On the lattice, Z2 symmetry makes the magnetic coupling multiplicatively renormalized: blocking an ergodic Z2-symmetric ensemble generates no odd coupling, the linearized RG is block-diagonal, and the magnetic eigenvalue $y_h = 15/8$ is reproduced. The thermal operator, the lattice counterpart of a scalar mass term, is relevant but unprotected: its critical value depends on every even coupling and is shifted by fluctuations by an amount of order the cutoff, so approaching criticality costs a tuning that grows like ξ/a . This is the structure of the hierarchy problem, in a model where every step can be computed. Among lattice mechanisms, Kramers–Wannier self-duality protects the even operator; it shares several formal features with goofy transformations and, unlike them, survives a positive cutoff. Whether goofy transformations protect the Higgs mass in a physically regulated theory remains open: at one loop with a hard cutoff, they do not protect the 2HDM mass relation unless the regulator itself transforms.

14. Proposed final thesis structure

One argument, in order: what naturalness predicts; the lattice test of both halves of the prediction; the QFT calculation of both halves; and what goofy would have to add. The main text stays close to its current length (about 25–30 pages), with the new technical material in appendices.

#	Chapter	Content	Figures	Status
1	Introduction	The hierarchy problem as a naturalness question; the precise question (protected versus unprotected relevant operators in one RG); the lattice-QFT dictionary; what the thesis shows and what it does not	—	Revised
1.1	RG essentials	Eigenvalues γ and relevance; scaling fields; symmetries as invariant hyperplanes, with the regulator and anomaly caveats; the tuning measure $\Delta \sim (\xi/a)^\gamma$	—	Revised
2	The protected direction on the lattice	Model, Z2-covariant blocking, inference with errors; T1 and T2; the null test and the single-sign leakage; the MCRG odd block and γ_h	Traces; odd couplings against b ; MCRG	Rewritten
3	The unprotected direction on the lattice	Thermal scaling field and γ_t ; the critical line against mean field; the tuning cost at the thesis point; the contrast with h	Binder test; critical line	New
4	Mass corrections in QFT	Fermion and scalar self-energies in dimensional regularization (existing) and with a cutoff (T3); the dictionary to Chapters 2–3	Self-energy diagrams	Revised

#	Chapter	Content	Figures	Status
5	Goofy transformations	The 2HDM r_0 relations and their RG stability, attributed precisely; the SM extension and its open points; outer automorphisms; the regulator caveat (T4)	—	Revised
5.4	A lattice analogue: KW duality	Self-dual chain, exact diagonalization, parallels and differences with goofy, labelled as an analogy	KW crossings	New
6	Conclusions	The conclusion proposed in §13	—	Rewritten
A	Simulation and inference	Wolff with a field; validation suite; autocorrelation and error calibration; Newton optimizer; basis tables; what the original protocol did and why its outputs differ	Magnetization traces (if not in Ch. 2)	Revised
B	Proofs	T1 (symmetric data, any truncation) and T2 (single-sign leakage)	—	New
C	MCRG	Method, truncation and finite-size tables, scaling-field components	—	New
D	Exact diagonalization	Parity sectors, crossing method, extrapolation and its bias	—	New
E	Loop calculations	Existing B.1–B.2, cutoff integrals, the T4 calculation	—	Revised

Why this order. Chapters 2 and 3 make naturalness visible on both sides with the same pipeline, so the contrast in Chapter 4 refers back to something measured rather than asserted. Chapter 5 then asks of goofy exactly what Chapter 3 shows is needed — protection of the even, relevant operator — and §5.4 shows that such protection can exist on a lattice while making clear where goofy differs. The reader leaves with one claim per chapter and a clear line between demonstration and analogy.

15. Remaining work for a human researcher

Ranked by necessity. Time estimates assume the code on the branch extension-symmetry-protection, which is delivered as a git bundle because this session could not push; publishing it needs the author's credentials.

Before submission.

1. **Decide the scope of the lattice chapter.** Recommended: adopt the unprotected direction as a new §2.4 and rewrite §2.3 around the new figures (about a week of writing).
2. **Verify T4 independently.** Check the gauge and Yukawa parts of the 2HDM quadratic divergences against a published calculation in the thesis's conventions, and confirm that λ_5 – λ_7 do not enter the diagonal mass shifts (about two days).
3. **Read de Boer and Trautner (arXiv:2603.12318) in full** and align the wording on "symmetry of the beta functions versus symmetry of the action" with theirs (about a day).
4. **Re-check every literature statement in §4** against the sources in the §13 table, including loop orders and who showed what (about a day).
5. **Supervisor review of the KW-goofy framing.** It must read as an analogy, not as evidence.

Worth doing (a few days each).

6. **Null test at the critical point of the thesis model.** The data exist (8000 configurations at $K1 = 0.2059$, $L = 256$); the analysis is one command with `scripts/analyse_protection.py`.
7. **Error calibration.** Either inflate the odd-coupling errors by the factor measured in §10 or replace the linearized jackknife by a full delete-one refit.
8. **The field channel.** Measure the odd couplings at fixed sample magnetization for several h_0 with larger statistics and a sampler that handles the field efficiently (for example Swendsen–Wang with a ghost spin), or rely on MCRG for the field response, as this report does.

Extensions (weeks, optional).

9. **A two-component lattice analogue done right.** The self-dual line of the Ashkin–Teller model, simulated with a two-species cluster algorithm, would give a classical Monte Carlo counterpart of §9.7 and is the defensible form of candidate B.
10. **DMRG for the KW chains.** Chains of hundreds of sites would pin the self-dual transition to 10^{-5} and locate the tricritical endpoint of the protected line.
11. **A QFT-side follow-up to T4.** A two-loop or lattice-perturbation-theory check of the 2HDM mass relation with a physical regulator.

Not recommended. The complexified Ising model (candidate C) and goofy-like rotations of two Ising species without duality (candidate B as posed): the first has no positive measure, the second reduces to an ordinary Z4 that cannot forbid the even operator (§4).

Sources

Pages opened while preparing this report. Classic references recommended in §13 ('t Hooft 1980, Swendsen 1979, Wolff 1989, Kramers–Wannier 1941, Barbieri–Giudice 1988) are standard and were not re-opened; check their bibliographic details before citing. The page of Salas and Sokal (arXiv:cond-mat/9904038), the usual source for the exact critical Binder value 0.61069, could not be opened (rate limit) and should be checked too.

- [Repository: juanagranadosr/ising_odd_RG_flow](#)
- [L. Di Carlo, Monte Carlo, blocking, and inference: How to measure the renormalization group flow, arXiv:2401.04811](#)
- [P. M. Ferreira, B. Grzadkowski, O. M. Ogreid, P. Osland, New Symmetries of the Two-Higgs-Doublet Model, arXiv:2306.02410](#)
- [A. Trautner, Goofy is the new Normal, arXiv:2505.00099](#)
- [T. de Boer, F. Goertz, A. Incrocci, The goofy-symmetric Standard Model and the Hierarchy Problem, arXiv:2507.22111](#)
- [A. Trautner, Goofy transformations and the hierarchy problem, arXiv:2508.02646](#)
- [INSPIRE record of arXiv:2508.02646: Phys. Lett. B 873 \(2026\) 140190](#)
- [T. de Boer, A. Trautner, Outer automorphisms are sufficient conditions for RG fixed points, arXiv:2603.12318](#)
- [E. O'Brien, P. Fendley, Lattice supersymmetry and order-disorder coexistence in the tricritical Ising model, arXiv:1712.06662 \(PRL 120, 206403\)](#)
- [N. Seiberg, S. Seifnashri, S.-H. Shao, Non-invertible symmetries and LSM-type constraints on a tensor product Hilbert space, arXiv:2401.12281](#)
- [N. Darvishi, M. Krawczyk, Implication of Quadratic Divergences Cancellation in the Two Higgs Doublet Model, arXiv:1709.07219](#)